

```
In [1]: !pip install torch torchvision fiftyone albumentations opencv-python-headless tq
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: torch in /home/jupyter-st125990/.local/lib/python3.12/site-packages (2.6.0+cu124)
Requirement already satisfied: torchvision in /home/jupyter-st125990/.local/lib/python3.12/site-packages (0.21.0+cu124)
Collecting fiftyone
 Downloading fiftyone-1.17.0-py3-none-any.whl.metadata (22 kB)
Collecting albumentations
 Downloading albumentations-2.0.8-py3-none-any.whl.metadata (43 kB)
Collecting opencv-python-headless
 Downloading opencv_python_headless-4.13.0.92-cp37-abi3-manylinux_2_28_x86_64.whl.metadata (19 kB)
Requirement already satisfied: tqdm in /opt/tljh/user/lib/python3.12/site-packages (4.66.5)
Collecting pycocotools
 Downloading pycocotools-2.0.11-cp312-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (1.3 kB)
Collecting torchmetrics
 Downloading torchmetrics-1.9.0-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: filelock in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (3.29.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (4.15.0)
Requirement already satisfied: networkx in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (2026.4.0)
Requirement already satisfied: NVIDIA-CUDA-nvrtc-cu12==12.4.127 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (12.4.127)
Requirement already satisfied: NVIDIA-CUDA-runtime-cu12==12.4.127 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (12.4.127)
Requirement already satisfied: NVIDIA-CUDA-cupti-cu12==12.4.127 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (12.4.127)
Requirement already satisfied: NVIDIA-CUDNN-cu12==9.1.0.70 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (9.1.0.70)
Requirement already satisfied: NVIDIA-CUBLAS-cu12==12.4.5.8 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (12.4.5.8)
Requirement already satisfied: NVIDIA-CUFFT-cu12==11.2.1.3 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (11.2.1.3)
Requirement already satisfied: NVIDIA-CURAND-cu12==10.3.5.147 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (10.3.5.147)
Requirement already satisfied: NVIDIA-CUSOLVER-cu12==11.6.1.9 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (11.6.1.9)
Requirement already satisfied: NVIDIA-CUSPARSE-cu12==12.3.1.170 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (12.3.1.170)
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Requirement already satisfied: triton==3.2.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (3.2.0)
Requirement already satisfied: setuptools in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torch) (70.2.0)
Requirement already satisfied: sympy==1.13.1 in /home/jupyter-st125990/.local/lib

b/python3.12/site-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from sympy==1.13.1->torch) (1.3.0)
Requirement already satisfied: numpy in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torchvision) (2.4.4)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from torchvision) (12.2.0)
Collecting Deprecated (from fiftyone)
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Requirement already satisfied: packaging in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (24.1)
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Collecting argcomplete<4,>=2 (from fiftyone)
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Requirement already satisfied: async_lru<3,>=2 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (2.0.4)
Requirement already satisfied: beautifulsoup4<5,>=2 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (4.12.3)
Collecting boto3<2,>=1 (from fiftyone)
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Collecting cachetools<7,>=5 (from fiftyone)
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Collecting dacite<2,>=1.6.0 (from fiftyone)
 Downloading dacite-1.9.2-py3-none-any.whl.metadata (17 kB)
Collecting dill<0.5,>=0.1 (from fiftyone)
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Collecting exceptiongroup<2,>=1 (from fiftyone)
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Collecting ftfy<7,>=4 (from fiftyone)
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Collecting humanize<5,>=2 (from fiftyone)
 Downloading humanize-4.15.0-py3-none-any.whl.metadata (7.8 kB)
Collecting hypercorn<0.19,>=0.13.2 (from fiftyone)
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Requirement already satisfied: jsonpatch<2,>=1 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (1.33)
Collecting mongoengine~=0.29.1 (from fiftyone)
 Downloading mongoengine-0.29.3-py3-none-any.whl.metadata (7.0 kB)
Collecting motor~=3.6.0 (from fiftyone)
 Downloading motor-3.6.1-py3-none-any.whl.metadata (21 kB)
Collecting plotly<7,>=6.1.1 (from fiftyone)
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Collecting pprintpp<0.5,>=0.1 (from fiftyone)
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Collecting protobuf==6.33.6 (from fiftyone)
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Requirement already satisfied: psutil<8,>=5 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (5.9.8)
Collecting pydash<9,>=6 (from fiftyone)
 Downloading pydash-8.0.6-py3-none-any.whl.metadata (3.4 kB)
Collecting pymongo~=4.9.2 (from fiftyone)
 Downloading pymongo-4.9.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (22 kB)
Requirement already satisfied: pytz in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (2024.2)
Requirement already satisfied: PyYAML<7,>=4 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (6.0.2)
Collecting regex (from fiftyone)

Downloading regex-2026.5.9-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40 kB)
 Collecting retrying<2,>=1 (from fiftyone)
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 Collecting sseclient-py<2,>=1.7.2 (from fiftyone)
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 Collecting sse-starlette<4,>=0.10.3 (from fiftyone)
 Downloading sse_starlette-3.4.4-py3-none-any.whl.metadata (15 kB)
 Collecting starlette<0.53,>=0.49.1 (from fiftyone)
 Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
 Collecting strawberry-graphql<0.317.0,>=0.312.3 (from fiftyone)
 Downloading strawberry_graphql-0.316.0-py3-none-any.whl.metadata (6.9 kB)
 Collecting tabulate<0.10,>=0.7 (from fiftyone)
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 Collecting xmldict<2,>=1 (from fiftyone)
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 Collecting universal-analytics-python3<2,>=1.0.1 (from fiftyone)
 Downloading universal_analytics_python3-1.1.1-py3-none-any.whl.metadata (5.5 kB)
 Requirement already satisfied: matplotlib<4 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from fiftyone) (3.10.9)
 Requirement already satisfied: pandas<4 in /opt/tljh/user/lib/python3.12/site-packages (from fiftyone) (2.2.3)
 Collecting rtree<2 (from fiftyone)
 Downloading rtree-1.4.1-py3-none-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (2.1 kB)
 Collecting scikit-learn<2 (from fiftyone)
 Downloading scikit_learn-1.9.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
 Collecting scikit-image<1 (from fiftyone)
 Downloading scikit_image-0.26.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (15 kB)
 Collecting scipy<2 (from fiftyone)
 Downloading scipy-1.17.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (62 kB)
 Collecting fiftyone-brain<0.23,>=0.22.0 (from fiftyone)
 Downloading fiftyone_brain-0.22.0-py3-none-any.whl.metadata (5.2 kB)
 Collecting fiftyone-db<2.0,>=0.4 (from fiftyone)
 Downloading fiftyone_db-1.4.1.tar.gz (8.6 kB)
 Installing build dependencies ... done
 Getting requirements to build wheel ... done
 Preparing metadata (pyproject.toml) ... done
 Collecting voxel51-eta<0.17,>=0.16.0 (from fiftyone)
 Downloading voxel51_eta-0.16.0-py2.py3-none-any.whl.metadata (15 kB)
 Requirement already satisfied: pydantic>=2.9.2 in /opt/tljh/user/lib/python3.12/site-packages (from albucentations) (2.10.4)
 Collecting albucentations==0.0.24 (from albucentations)
 Downloading albucentations-0.0.24-py3-none-any.whl.metadata (5.3 kB)
 Collecting stringzilla>=3.10.4 (from albucentations==0.0.24->albucentations)
 Downloading stringzilla-4.6.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (121 kB)
 Collecting simsimd>=5.9.2 (from albucentations==0.0.24->albucentations)
 Downloading simsimd-6.5.16-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (70 kB)
 Collecting lightning-utilities>=0.15.3 (from torchmetrics)
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 Requirement already satisfied: soupsieve>1.2 in /opt/tljh/user/lib/python3.12/site-packages (from beautifulsoup4<5,>=2->fiftyone) (2.6)
 Collecting boto3<1.44.0,>=1.43.26 (from boto3<2,>=1->fiftyone)
 Downloading boto3-1.43.26-py3-none-any.whl.metadata (5.6 kB)

Collecting jmespath<2.0.0,>=0.7.1 (from boto3<2,>=1->fiftyone)
 Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
 Collecting s3transfer<0.19.0,>=0.18.0 (from boto3<2,>=1->fiftyone)
 Downloading s3transfer-0.18.0-py3-none-any.whl.metadata (1.7 kB)
 Requirement already satisfied: wcwidth in /opt/tljh/user/lib/python3.12/site-packages (from ftfy<7,>=4->fiftyone) (0.2.13)
 Requirement already satisfied: h11 in /opt/tljh/user/lib/python3.12/site-packages (from hypercorn<0.19,>=0.13.2->fiftyone) (0.14.0)
 Collecting h2>=4.3.0 (from hypercorn<0.19,>=0.13.2->fiftyone)
 Downloading h2-4.3.0-py3-none-any.whl.metadata (5.1 kB)
 Collecting priority (from hypercorn<0.19,>=0.13.2->fiftyone)
 Downloading priority-2.0.0-py3-none-any.whl.metadata (6.6 kB)
 Collecting wsproto>=0.14.0 (from hypercorn<0.19,>=0.13.2->fiftyone)
 Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)
 Requirement already satisfied: MarkupSafe>=2.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from jinja2->torch) (3.0.3)
 Requirement already satisfied: jsonpointer>=1.9 in /opt/tljh/user/lib/python3.12/site-packages (from jsonpatch<2,>=1->fiftyone) (3.0.0)
 Requirement already satisfied: contourpy>=1.0.1 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (1.3.3)
 Requirement already satisfied: cycler>=0.10 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (4.63.0)
 Requirement already satisfied: kiwisolver>=1.3.1 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (1.5.0)
 Requirement already satisfied: pyparsing>=3 in /home/jupyter-st125990/.local/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (3.3.2)
 Requirement already satisfied: python-dateutil>=2.7 in /opt/tljh/user/lib/python3.12/site-packages (from matplotlib<4->fiftyone) (2.9.0.post0)
 Requirement already satisfied: tzdata>=2022.7 in /opt/tljh/user/lib/python3.12/site-packages (from pandas<4->fiftyone) (2024.2)
 Collecting narwhals>=1.15.1 (from plotly<7,>=6.1.1->fiftyone)
 Downloading narwhals-2.22.1-py3-none-any.whl.metadata (15 kB)
 Requirement already satisfied: annotated-types>=0.6.0 in /opt/tljh/user/lib/python3.12/site-packages (from pydantic>=2.9.2->alumentations) (0.7.0)
 Requirement already satisfied: pydantic-core==2.27.2 in /opt/tljh/user/lib/python3.12/site-packages (from pydantic>=2.9.2->alumentations) (2.27.2)
 Collecting dnspython<3.0.0,>=1.16.0 (from pymongo==4.9.2->fiftyone)
 Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
 Collecting imageio!=2.35.0,>=2.33 (from scikit-image<1->fiftyone)
 Downloading imageio-2.37.3-py3-none-any.whl.metadata (9.7 kB)
 Collecting tifffile>=2022.8.12 (from scikit-image<1->fiftyone)
 Downloading tifffile-2026.6.1-py3-none-any.whl.metadata (33 kB)
 Collecting lazy-loader>=0.4 (from scikit-image<1->fiftyone)
 Downloading lazy_loader-0.5-py3-none-any.whl.metadata (5.9 kB)
 Collecting joblib>=1.4.0 (from scikit-learn<2->fiftyone)
 Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
 Collecting threadpoolctl>=3.5.0 (from scikit-learn<2->fiftyone)
 Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
 Requirement already satisfied: anyio>=4.7.0 in /opt/tljh/user/lib/python3.12/site-packages (from sse-starlette<4,>=0.10.3->fiftyone) (4.7.0)
 Collecting graphql-core<3.4.0,>=3.2.0 (from strawberry-graphql<0.317.0,>=0.312.3->fiftyone)
 Downloading graphql_core-3.2.11-py3-none-any.whl.metadata (11 kB)
 Collecting cross-web>=0.6.0 (from strawberry-graphql<0.317.0,>=0.312.3->fiftyone)
 Downloading cross_web-0.7.0-py3-none-any.whl.metadata (5.1 kB)
 Requirement already satisfied: httpx>=0.10.0 in /opt/tljh/user/lib/python3.12/site-packages (from universal-analytics-python<2,>=1.0.1->fiftyone) (0.28.1)
 Collecting glob2 (from voxel51-eta<0.17,>=0.16.0->fiftyone)

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    Downloading glob2-0.7.tar.gz (10 kB)
    Preparing metadata (setup.py) ... done
Collecting jsonlines (from voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading jsonlines-4.0.0-py3-none-any.whl.metadata (1.6 kB)
Collecting paramiko<4,>=3 (from voxel51-eta<0.17,>=0.16.0->fiftyone)
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Collecting py7zr (from voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading py7zr-1.1.0-py3-none-any.whl.metadata (17 kB)
Collecting rarfile (from voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading rarfile-4.2-py3-none-any.whl.metadata (4.4 kB)
Requirement already satisfied: requests in /opt/tljh/user/lib/python3.12/site-packages (from voxel51-eta<0.17,>=0.16.0->fiftyone) (2.32.3)
Collecting sortedcontainers (from voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting tzlocal (from voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading tzlocal-5.3.1-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: urllib3 in /opt/tljh/user/lib/python3.12/site-packages (from voxel51-eta<0.17,>=0.16.0->fiftyone) (2.2.3)
Collecting wrapt<3,>=1.10 (from Deprecated->fiftyone)
    Downloading wrapt-2.2.1-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.4 kB)
Requirement already satisfied: idna>=2.8 in /opt/tljh/user/lib/python3.12/site-packages (from anyio>=4.7.0->sse-starlette<4,>=0.10.3->fiftyone) (3.10)
Requirement already satisfied: sniffio>=1.1 in /opt/tljh/user/lib/python3.12/site-packages (from anyio>=4.7.0->sse-starlette<4,>=0.10.3->fiftyone) (1.3.1)
Collecting hyperframe<7,>=6.1 (from h2>=4.3.0->hypercorn<0.19,>=0.13.2->fiftyone)
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Collecting hpack<5,>=4.1 (from h2>=4.3.0->hypercorn<0.19,>=0.13.2->fiftyone)
    Downloading hpack-4.1.0-py3-none-any.whl.metadata (4.6 kB)
Requirement already satisfied: certifi in /opt/tljh/user/lib/python3.12/site-packages (from httpx>=0.10.0->universal-analytics-python3<2,>=1.0.1->fiftyone) (2024.12.14)
Requirement already satisfied: httpcore==1.* in /opt/tljh/user/lib/python3.12/site-packages (from httpx>=0.10.0->universal-analytics-python3<2,>=1.0.1->fiftyone) (1.0.7)
Collecting bcrypt>=3.2 (from paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: cryptography>=3.3 in /opt/tljh/user/lib/python3.12/site-packages (from paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone) (44.0.0)
Collecting pynacl>=1.5 (from paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading pynacl-1.6.2-cp38-abi3-manylinux_2_34_x86_64.whl.metadata (10.0 kB)
Requirement already satisfied: six>=1.5 in /opt/tljh/user/lib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib<4->fiftyone) (1.17.0)
INFO: pip is looking at multiple versions of wsproto to determine which version is compatible with other requirements. This could take a while.
Collecting wsproto>=0.14.0 (from hypercorn<0.19,>=0.13.2->fiftyone)
    Downloading wsproto-1.3.1-py3-none-any.whl.metadata (5.2 kB)
    Downloading wsproto-1.3.0-py3-none-any.whl.metadata (5.2 kB)
    Downloading wsproto-1.2.0-py3-none-any.whl.metadata (5.6 kB)
Requirement already satisfied: attrs>=19.2.0 in /opt/tljh/user/lib/python3.12/site-packages (from jsonlines->voxel51-eta<0.17,>=0.16.0->fiftyone) (24.3.0)
Collecting texttable (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)
Collecting pycryptodomex>=3.20.0 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading pycryptodomex-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.4 kB)
Collecting brotli>=1.2.0 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
    Downloading brotli-1.2.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_6

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4.whl.metadata (6.1 kB)
Collecting backports.zstd>=1.0.0 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading backports_zstd-1.5.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (6.8 kB)
Collecting pyppmd>=1.3.1 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading pyppmd-1.3.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (5.4 kB)
Collecting pybcj>=1.0.6 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading pybcj-1.0.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (3.9 kB)
Collecting multivolumefile>=0.2.3 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading multivolumefile-0.2.3-py3-none-any.whl.metadata (6.3 kB)
Collecting inflate64>=1.0.4 (from py7zr->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading inflate64-1.0.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (4.4 kB)
Requirement already satisfied: charset-normalizer<4,>=2 in /opt/tljh/user/lib/python3.12/site-packages (from requests->voxel51-eta<0.17,>=0.16.0->fiftyone) (3.3.2)
Requirement already satisfied: cffi>=1.12 in /opt/tljh/user/lib/python3.12/site-packages (from cryptography>=3.3->paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone) (1.17.1)
Collecting cffi>=1.12 (from cryptography>=3.3->paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone)
 Downloading cffi-2.0.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.6 kB)
Requirement already satisfied: pycparser in /opt/tljh/user/lib/python3.12/site-packages (from cffi>=1.12->cryptography>=3.3->paramiko<4,>=3->voxel51-eta<0.17,>=0.16.0->fiftyone) (2.22)
Downloading fiftyone-1.17.0-py3-none-any.whl (17.8 MB)
----- 17.8/17.8 MB 22.7 MB/s eta 0:00:000:010:01
Downloading protobuf-6.33.6-cp39-abi3-manylinux2014_x86_64.whl (323 kB)
Downloading alumentations-2.0.8-py3-none-any.whl (369 kB)
Downloading albucore-0.0.24-py3-none-any.whl (15 kB)
Downloading opencv_python_headless-4.13.0.92-cp37-abi3-manylinux_2_28_x86_64.whl (60.4 MB)
----- 60.4/60.4 MB 25.1 MB/s eta 0:00:000:0100:01
Downloading pycocotools-2.0.11-cp312-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (411 kB)
Downloading torchmetrics-1.9.0-py3-none-any.whl (983 kB)
----- 983.4/983.4 kB 18.0 MB/s eta 0:00:00
Downloading aiofiles-25.1.0-py3-none-any.whl (14 kB)
Downloading argcomplete-3.6.3-py3-none-any.whl (43 kB)
Downloading boto3-1.43.26-py3-none-any.whl (140 kB)
Downloading cachetools-6.2.6-py3-none-any.whl (11 kB)
Downloading dacite-1.9.2-py3-none-any.whl (16 kB)
Downloading dill-0.4.1-py3-none-any.whl (120 kB)
Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Downloading fiftyone_brain-0.22.0-py3-none-any.whl (112 kB)
Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
Downloading humanize-4.15.0-py3-none-any.whl (132 kB)
Downloading hypercorn-0.18.0-py3-none-any.whl (61 kB)
Downloading lightning_utilities-0.15.3-py3-none-any.whl (31 kB)
Downloading mongoengine-0.29.3-py3-none-any.whl (112 kB)
Downloading motor-3.6.1-py3-none-any.whl (74 kB)
Downloading plotly-6.8.0-py3-none-any.whl (9.9 MB)
----- 9.9/9.9 MB 21.4 MB/s eta 0:00:0000:01

```
Downloading pprintpp-0.4.0-py2.py3-none-any.whl (16 kB)
Downloading pydash-8.0.6-py3-none-any.whl (101 kB)
Downloading pymongo-4.9.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.9 MB)
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Downloading retrying-1.4.2-py3-none-any.whl (10 kB)
Downloading rtree-1.4.1-py3-none-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (507 kB)
Downloading scikit_image-0.26.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (13.6 MB)
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010:01
Downloading scikit_learn-1.9.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (9.1 MB)
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Downloading scipy-1.17.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (35.2 MB)
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Downloading sse_starlette-3.4.4-py3-none-any.whl (16 kB)
Downloading sseclient_py-1.9.0-py3-none-any.whl (8.4 kB)
Downloading starlette-0.52.1-py3-none-any.whl (74 kB)
Downloading strawberry_graphql-0.316.0-py3-none-any.whl (326 kB)
Downloading tabulate-0.9.0-py3-none-any.whl (35 kB)
Downloading universal_analytics_python3-1.1.1-py3-none-any.whl (10 kB)
Downloading voxel51_eta-0.16.0-py2.py3-none-any.whl (935 kB)
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Downloading xmltodict-1.0.4-py3-none-any.whl (13 kB)
Downloading deprecated-1.3.1-py2.py3-none-any.whl (11 kB)
Downloading regex-2026.5.9-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (801 kB)
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Downloading boto3-1.43.26-py3-none-any.whl (15.2 MB)
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Downloading cross_web-0.7.0-py3-none-any.whl (25 kB)
Downloading dnspython-2.8.0-py3-none-any.whl (331 kB)
Downloading graphql_core-3.2.11-py3-none-any.whl (214 kB)
Downloading h2-4.3.0-py3-none-any.whl (61 kB)
Downloading imageio-2.37.3-py3-none-any.whl (317 kB)
Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
Downloading lazy_loader-0.5-py3-none-any.whl (8.0 kB)
Downloading narwhals-2.22.1-py3-none-any.whl (454 kB)
Downloading paramiko-3.5.1-py3-none-any.whl (227 kB)
Downloading s3transfer-0.18.0-py3-none-any.whl (88 kB)
Downloading simsimd-6.5.16-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (583 kB)
_____ 583.5/583.5 kB 12.5 MB/s eta 0:00:00
Downloading stringzilla-4.6.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (1.9 MB)
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Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Downloading tifffile-2026.6.1-py3-none-any.whl (267 kB)
Downloading wrapt-2.2.1-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (169 kB)
Downloading wsproto-1.2.0-py3-none-any.whl (24 kB)
Downloading jsonlines-4.0.0-py3-none-any.whl (8.7 kB)
Downloading priority-2.0.0-py3-none-any.whl (8.9 kB)
Downloading py7zr-1.1.0-py3-none-any.whl (71 kB)
```

Downloading rarfile-4.2-py3-none-any.whl (29 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Downloading tzlocal-5.3.1-py3-none-any.whl (18 kB)
Downloading backports_zstd-1.5.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (495 kB)
Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
Downloading brotli-1.2.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.4 MB)

1.4/1.4 MB 22.6 MB/s eta 0:00:00

Downloading hpack-4.1.0-py3-none-any.whl (34 kB)
Downloading hyperframe-6.1.0-py3-none-any.whl (13 kB)
Downloading inflate64-1.0.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (100 kB)
Downloading multivolumefile-0.2.3-py3-none-any.whl (17 kB)
Downloading pybcj-1.0.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (51 kB)
Downloading pycryptodomex-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)

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Downloading pynacl-1.6.2-cp38-abi3-manylinux_2_34_x86_64.whl (1.4 MB)

1.4/1.4 MB 24.0 MB/s eta 0:00:00

Downloading pyppmd-1.3.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (144 kB)
Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)
Downloading cffi-2.0.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (219 kB)

Building wheels for collected packages: fiftyone-db, glob2

Building wheel for fiftyone-db (pyproject.toml) ... one

Created wheel for fiftyone-db: filename=fiftyone_db-1.4.1-py3-none-manylinux1_x86_64.whl size=59387759 sha256=0ca834cee6aab6aac0bbbd8b6522b86c7c2f2ed43840e9bce5e6caf34fe0198d

Stored in directory: /home/jupyter-st125990/.cache/pip/wheels/fc/06/06/0eb7530b99c4934bf36390769cff3f0052fd04f50f390d5c8b

Building wheel for glob2 (setup.py) ... one

Created wheel for glob2: filename=glob2-0.7-py2.py3-none-any.whl size=9301 sha256=5b2ee7447c1dfb28c2fbf01f072a1257946d2318f68eaaa1995d35116944b26a

Stored in directory: /home/jupyter-st125990/.cache/pip/wheels/fb/b3/7b/71827dcd17a71be7a8b7adf9f30a4200b612a903d7d8af3442

Successfully built fiftyone-db glob2

Installing collected packages: texttable, sortedcontainers, simsimd, pprintpp, glob2, brotli, xmldict, wsproto, wrapt, tzlocal, tiffiff, threadpoolctl, tabulate, stringzilla, sseclient-py, scipy, rtree, retrying, regex, rarfile, pyppmd, pydash, pycryptodomex, pycocotools, pybcj, protobuf, priority, opencv-python-headless, narwhals, multivolumefile, lightning-utilities, lazy-loader, jsonlines, joblib, jmespath, inflate64, imageio, hyperframe, humanize, hpack, graphql-core, ftfy, fiftyone-db, exceptiongroup, dnspython, dill, dacite, cross-web, cffi, cachetools, bcrypt, backports.zstd, argcomplete, aiofiles, strawberry-graphql, starlette, scikit-learn, scikit-image, pynacl, pymongo, py7zr, plotly, h2, Deprecated, boto3, albucore, universal-analytics-python3, sse-starlette, s3transfer, paramiko, motor, mongoengine, hypercorn, fiftyone-brain, alumentations, voxel51-eta, torchmetrics, boto3, fiftyone

WARNING: The script pypprint is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts lsm2bin, tiff2fsspec, tiffcomment and tiffiff are installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script tabulate is installed in '/home/jupyter-st125990/.local/bin'

n' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts sz_split and sz_wc are installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts imageio_download_bin and imageio_remove_bin are installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script ftfy is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The scripts activate-global-python-argcomplete, python-argcomplete-check-easy-install-script and register-python-argcomplete are installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script strawberry is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script py7zr is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script plotly_get_chrome is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script hypercorn is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script eta is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script fiftyone is installed in '/home/jupyter-st125990/.local/bin' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

Successfully installed Deprecated-1.3.1 aiofiles-25.1.0 albucore-0.0.24 albumations-2.0.8 argcomplete-3.6.3 backports.zstd-1.5.0 bcrypt-5.0.0 boto3-1.43.26 botocore-1.43.26 brotli-1.2.0 cachetools-6.2.6 cffi-2.0.0 cross-web-0.7.0 dacite-1.9.2 dill-0.4.1 dnspython-2.8.0 exceptiongroup-1.3.1 fiftyone-1.17.0 fiftyone-brain-0.22.0 fiftyone-db-1.4.1 ftfy-6.3.1 glob2-0.7 graphql-core-3.2.11 h2-4.3.0 hpacak-4.1.0 humanize-4.15.0 hypercorn-0.18.0 hyperframe-6.1.0 imageio-2.37.3 inflate64-1.0.4 jmespath-1.1.0 joblib-1.5.3 jsonlines-4.0.0 lazy-loader-0.5 lightning-utilities-0.15.3 mongoengine-0.29.3 motor-3.6.1 multivolume-0.2.3 narwhals-2.22.1 opencv-python-headless-4.13.0.92 paramiko-3.5.1 plotly-6.8.0 pprintpp-0.4.0 priority-2.0.0 protobuf-6.33.6 py7zr-1.1.0 pybcj-1.0.7 pycocotools-2.0.11 pycryptodomex-3.23.0 pydash-8.0.6 pymongo-4.9.2 pynacl-1.6.2 pyppmd-1.3.1 rarfile-4.2 regex-2026.5.9 retrying-1.4.2 rtree-1.4.1 s3transfer-0.18.0 scikit-image-0.26.0 scikit-learn-1.9.0 scipy-1.17.1 simsimd-6.5.16 sortedcontainers-2.4.0 sse-starlette-3.4.4 sseclient-py-1.9.0 starlette-0.52.1 strawberry-graphql-0.316.0 stringzilla-4.6.2 tabulate-0.9.0 texttable-1.7.0 threadpoolctl-3.6.0 tiffio-2026.6.1 torchmet

rics-1.9.0 tzlocal-5.3.1 universal-analytics-python3-1.1.1 voxel51-eta-0.16.0 wrapt-2.2.1 wsproto-1.2.0 xmltodict-1.0.4

In [1]: !wget -q -nc https://github.com/ayooshkathuria/pytorch-yolo-v3/raw/master/dog-cy

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches
import torch
import torch.nn as nn
import torch.nn.functional as F

device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
```

Using device: cuda:0

GPU: NVIDIA GeForce RTX 2080 Ti

In [2]: *#Download COCO dataset*

```
import fiftyone as fo
import fiftyone.zoo as foz

# List available zoo datasets
print(foz.list_zoo_datasets())

# Download the COCO-2017 validation split and load it into FiftyOne
dataset = foz.load_zoo_dataset("coco-2017", split="validation")
dataset.persistent = True
```

['activitynet-100', 'activitynet-200', 'bdd100k', 'caltech101', 'caltech256', 'cifar10', 'cifar100', 'cityscapes', 'coco-2014', 'coco-2017', 'fashion-mnist', 'fiw', 'hmdb51', 'imagenet-2012', 'imagenet-sample', 'kinetics-400', 'kinetics-600', 'kinetics-700', 'kinetics-700-2020', 'kitti', 'kitti-multiview', 'lfw', 'mnist', 'open-images-v6', 'open-images-v7', 'places', 'quickstart', 'quickstart-3d', 'quickstart-geo', 'quickstart-groups', 'quickstart-video', 'sama-coco', 'ucf101', 'voc-2007', 'voc-2012']

Downloading split 'validation' to '/home/jupyter-st125990/fiftyone/coco-2017/validation' if necessary

Found annotations at '/home/jupyter-st125990/fiftyone/coco-2017/raw/instances_val2017.json'

Images already downloaded

Existing download of split 'validation' is sufficient

Loading existing dataset 'coco-2017-validation'. To reload from disk, either delete the existing dataset or provide a custom `dataset\_name` to use

In [3]: *#Download darknet.py and util.py*

```
import os
if not os.path.exists('darknet.py'):
    os.system('wget -q https://raw.githubusercontent.com/ayooshkathuria/YOLO_v3_
if not os.path.exists('util.py'):
    os.system('wget -q https://raw.githubusercontent.com/ayooshkathuria/YOLO_v3_
```

In [4]: *#Download YOLOv3.cfg*

```
import os
```

```

os.makedirs('cfg', exist_ok=True)
if not os.path.exists('cfg/yolov3.cfg'):
    os.system('wget -q https://raw.githubusercontent.com/ayooshkathuria/YOLO_v3_

```

In [5]: *#Set Dataset paths*

```

import os

fo_base = os.path.expanduser('~/.fiftyone/coco-2017')
path2data = os.path.join(fo_base, 'validation', 'data')
path2json = os.path.join(fo_base, 'raw', 'instances_val2017.json')

assert os.path.isdir(path2data), f'Images not found: {path2data}'
assert os.path.isfile(path2json), f'Annotations not found: {path2json}'

print(f'Images      : {path2data}')
print(f'Annotations: {path2json}')

img_size = 416

```

Images : /home/jupyter-st125990/fiftyone/coco-2017/validation/data
Annotations: /home/jupyter-st125990/fiftyone/coco-2017/raw/instances_val2017.json

In [6]: *# Define CustomCoco dataset class*

```

# coding=utf-8
import os, sys, math, cv2, random, json
import numpy as np
import torch
from PIL import Image
from torchvision.datasets import CocoDetection
from typing import Any, Callable, Optional, Tuple

ANCHORS = [
    [[10, 13], [16, 30], [33, 23]],
    [[30, 61], [62, 45], [59, 119]],
    [[116, 90], [156, 198], [373, 326]],
]
STRIDES = [8, 16, 32]
IP_SIZE = img_size
NUM_ANCHORS = 3
NUM_CLASSES = 80

with open(path2json) as js:
    data = json.load(js)["categories"]

cats_dict = {}
for i, cat in enumerate(data[:80]):
    cats_dict[str(cat['id'])] = i

class CustomCoco(CocoDetection):
    def __init__(self, root, annFile, transform=None,
                 target_transform=None, transforms=None):
        super(CocoDetection, self).__init__(root, transforms, transform, target_
        from pycocotools.coco import COCO
        self.coco = COCO(annFile)
        self.ids = list(sorted(self.coco.imgs.keys()))

    def __getitem__(self, index):

```

```

coco    = self.coco
img_id  = self.ids[index]
ann_ids = coco.getAnnIds(imgIds=img_id)
target  = coco.loadAnns(ann_ids)
path    = coco.loadImgs(img_id)[0]['file_name']

img = Image.open(os.path.join(self.root, path)).convert('RGB')
img = np.array(img)

category_ids = [obj['category_id'] for obj in target]
bboxes       = [obj['bbox']       for obj in target]

if self.transform is not None:
    transformed = self.transform(image=img, bboxes=bboxes, category_ids=category_ids)
    img         = transformed['image'],
    bboxes      = torch.Tensor(transformed['bboxes'])
    cat_ids     = torch.Tensor(transformed['category_ids'])
    labels, bboxes = self.__create_label(bboxes, cat_ids.type(torch.IntT

return img, labels, bboxes

def __len__(self):
    return len(self.ids)

def __create_label(self, bboxes, class_inds):
    bboxes      = np.array(bboxes)
    class_inds  = np.array(class_inds)
    strides     = np.array(STRIDES)
    train_output_size = IP_SIZE / strides
    anchors_per_scale = NUM_ANCHORS

    label = [
        np.zeros((int(train_output_size[i]), int(train_output_size[i]),
                  anchors_per_scale, 5 + NUM_CLASSES))
        for i in range(3)
    ]
    bboxes_xywh = [np.zeros((150, 4)) for _ in range(3)]
    bbox_count  = np.zeros((3,))

    for i in range(len(bboxes)):
        bbox_coor      = bboxes[i][:4]
        bbox_class_ind = cats_dict[str(class_inds[i])]

        one_hot = np.zeros(NUM_CLASSES, dtype=np.float32)
        one_hot[bbox_class_ind] = 1.0

        bbox_xywh = np.concatenate(
            [(0.5 * bbox_coor[2:] + bbox_coor[:2]), bbox_coor[2:]], axis=-1)
        bbox_xywh_scaled = 1.0 * bbox_xywh[np.newaxis, :] / strides[:, np.ne

    iou = []
    exist_positive = False
    for i in range(3):
        anchors_xywh = np.zeros((anchors_per_scale, 4))
        anchors_xywh[:, 0:2] = np.floor(bbox_xywh_scaled[i, 0:2]).astype
        anchors_xywh[:, 2:4] = ANCHORS[i]

        iou_scale = iou_xywh_numpy(bbox_xywh_scaled[i][np.newaxis, :], a
        iou.append(iou_scale)
        iou_mask = iou_scale > 0.3

```

```

if np.any(iou_mask):
    xind, yind = np.floor(bbox_xywh_scaled[i, 0:2]).astype(np.int)
    label[i][yind, xind, iou_mask, 0:4] = bbox_xywh * strides[i]
    label[i][yind, xind, iou_mask, 4:5] = 1.0
    label[i][yind, xind, iou_mask, 5:] = one_hot
    bbox_ind = int(bbox_count[i] % 150)
    bboxes_xywh[i][bbox_ind, :4] = bbox_xywh * strides[i]
    bbox_count[i] += 1
    exist_positive = True

if not exist_positive:
    best_anchor_ind = np.argmax(np.array(iou).reshape(-1), axis=-1)
    best_detect = int(best_anchor_ind / anchors_per_scale)
    best_anchor = int(best_anchor_ind % anchors_per_scale)
    xind, yind = np.floor(bbox_xywh_scaled[best_detect, 0:2]).astype(np.int)
    label[best_detect][yind, xind, best_anchor, 0:4] = bbox_xywh * strides[best_detect]
    label[best_detect][yind, xind, best_anchor, 4:5] = 1.0
    label[best_detect][yind, xind, best_anchor, 5:] = one_hot
    bbox_ind = int(bbox_count[best_detect] % 150)
    bboxes_xywh[best_detect][bbox_ind, :4] = bbox_xywh * strides[best_detect]
    bbox_count[best_detect] += 1

fs = int(train_output_size[2]) * int(train_output_size[2]) * anchors_per_scale
fm = int(train_output_size[1]) * int(train_output_size[1]) * anchors_per_scale
fl = int(train_output_size[0]) * int(train_output_size[0]) * anchors_per_scale

label_s = torch.Tensor(label[2]).view(1, fs, 5 + NUM_CLASSES).squeeze(0)
label_m = torch.Tensor(label[1]).view(1, fm, 5 + NUM_CLASSES).squeeze(0)
label_l = torch.Tensor(label[0]).view(1, fl, 5 + NUM_CLASSES).squeeze(0)

bboxes_s = torch.Tensor(bboxes_xywh[2])
bboxes_m = torch.Tensor(bboxes_xywh[1])
bboxes_l = torch.Tensor(bboxes_xywh[0])

labels = torch.cat([label_l, label_m, label_s], 0)
bboxes = torch.cat([bboxes_l, bboxes_m, bboxes_s], 0)
return labels, bboxes

```

In [7]: *#Define iou_xywh_numpy and CIOU_xywh_torch*

```

def iou_xywh_numpy(boxes1, boxes2):
    boxes1 = np.array(boxes1)
    boxes2 = np.array(boxes2)

    boxes1_area = boxes1[..., 2] * boxes1[..., 3]
    boxes2_area = boxes2[..., 2] * boxes2[..., 3]

    boxes1 = np.concatenate([boxes1[..., :2] - boxes1[..., 2:] * 0.5,
                              boxes1[..., :2] + boxes1[..., 2:] * 0.5], axis=-1)
    boxes2 = np.concatenate([boxes2[..., :2] - boxes2[..., 2:] * 0.5,
                              boxes2[..., :2] + boxes2[..., 2:] * 0.5], axis=-1)

    left_up = np.maximum(boxes1[..., :2], boxes2[..., :2])
    right_down = np.minimum(boxes1[..., 2:], boxes2[..., 2:])

    inter_section = np.maximum(right_down - left_up, 0.0)
    inter_area = inter_section[..., 0] * inter_section[..., 1]
    union_area = boxes1_area + boxes2_area - inter_area
    return 1.0 * inter_area / union_area

```

```

def CIoU_xywh_torch(bboxes1, bboxes2):
    bboxes1 = torch.cat([bboxes1[..., :2] - bboxes1[..., 2:] * 0.5,
                          bboxes1[..., :2] + bboxes1[..., 2:] * 0.5], dim=-1)
    bboxes2 = torch.cat([bboxes2[..., :2] - bboxes2[..., 2:] * 0.5,
                          bboxes2[..., :2] + bboxes2[..., 2:] * 0.5], dim=-1)

    bboxes1 = torch.cat([torch.min(bboxes1[..., :2], bboxes1[..., 2:]),
                          torch.max(bboxes1[..., :2], bboxes1[..., 2:])], dim=-1)
    bboxes2 = torch.cat([torch.min(bboxes2[..., :2], bboxes2[..., 2:]),
                          torch.max(bboxes2[..., :2], bboxes2[..., 2:])], dim=-1)

    bboxes1_area = (bboxes1[..., 2] - bboxes1[..., 0]) * (bboxes1[..., 3] - bboxes1[.
    bboxes2_area = (bboxes2[..., 2] - bboxes2[..., 0]) * (bboxes2[..., 3] - bboxes2[.

    inter_left_up    = torch.max(bboxes1[..., :2], bboxes2[..., :2])
    inter_right_down = torch.min(bboxes1[..., 2:], bboxes2[..., 2:])
    inter_section     = torch.max(inter_right_down - inter_left_up,
                                  torch.zeros_like(inter_right_down))
    inter_area        = inter_section[..., 0] * inter_section[..., 1]
    union_area        = bboxes1_area + bboxes2_area - inter_area
    ious              = 1.0 * inter_area / union_area

    outer_left_up     = torch.min(bboxes1[..., :2], bboxes2[..., :2])
    outer_right_down  = torch.max(bboxes1[..., 2:], bboxes2[..., 2:])
    outer             = torch.max(outer_right_down - outer_left_up,
                                  torch.zeros_like(inter_right_down))
    outer_diagonal_line = torch.pow(outer[..., 0], 2) + torch.pow(outer[..., 1],

    bboxes1_center = (bboxes1[..., :2] + bboxes1[..., 2:]) * 0.5
    bboxes2_center = (bboxes2[..., :2] + bboxes2[..., 2:]) * 0.5
    center_dis      = (torch.pow(bboxes1_center[..., 0] - bboxes2_center[..., 0], 2) +
                       torch.pow(bboxes1_center[..., 1] - bboxes2_center[..., 1], 2))

    bboxes1_size = torch.max(bboxes1[..., 2:] - bboxes1[..., :2],
                              torch.zeros_like(inter_right_down))
    bboxes2_size = torch.max(bboxes2[..., 2:] - bboxes2[..., :2],
                              torch.zeros_like(inter_right_down))
    v = (4 / (math.pi ** 2)) * torch.pow(
        torch.atan(bboxes1_size[..., 0] / torch.clamp(bboxes1_size[..., 1], min=1e
        torch.atan(bboxes2_size[..., 0] / torch.clamp(bboxes2_size[..., 1], min=1e

    alpha = v / (1 - ious + v)
    return ious - (center_dis / outer_diagonal_line + alpha * v)

```

In [8]: # Training Imports

```

from __future__ import division
import time, os, os.path as osp, random
import numpy as np, cv2
import pickle as pkl
import pandas as pd
import matplotlib.pyplot as plt
from copy import copy, deepcopy

import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim

```

```

from torch.autograd import Variable
from torch.utils.data import Subset

import torchvision
from torchvision import datasets, models, transforms

from util import *
import albumentations as A

```

In [9]: *# Define compute_map()*

```

from tqdm.auto import tqdm
import warnings, time

def compute_map(model, dataloader, device, img_size,
                conf_thresh=0.05, nms_thresh=0.4, num_classes=80):
    """Compute mAP@0.5 over the dataloader using torchmetrics."""
    try:
        from torchmetrics.detection import MeanAveragePrecision
        from torchvision.ops import nms as tv_nms
    except ImportError:
        print("torchmetrics/torchvision not installed - skipping mAP")
        return None

    warnings.filterwarnings("ignore", message="Encountered more than")
    metric = MeanAveragePrecision(
        iou_type="bbox",
        iou_thresholds=[0.5],
        max_detection_thresholds=[1, 10, 100],
    )
    model.eval()

    with torch.no_grad(), warnings.catch_warnings():
        warnings.simplefilter("ignore")
        for inputs, labels, bboxes in tqdm(dataloader, desc=" mAP eval", leave=
            inputs = torch.from_numpy(np.array(inputs)).squeeze(1).permute(0,3,1
            use_cuda = (device.type == "cuda")
            outputs = model(inputs, use_cuda)

            pred_conf = torch.sigmoid(outputs[..., 4])
            pred_cls = torch.sigmoid(outputs[..., 5:])
            pred_xy = outputs[..., 0:2]
            pred_wh = outputs[..., 2:4]

            pred_x1 = (pred_xy[..., 0] - pred_wh[..., 0] / 2).clamp(0, img_size)
            pred_y1 = (pred_xy[..., 1] - pred_wh[..., 1] / 2).clamp(0, img_size)
            pred_x2 = (pred_xy[..., 0] + pred_wh[..., 0] / 2).clamp(0, img_size)
            pred_y2 = (pred_xy[..., 1] + pred_wh[..., 1] / 2).clamp(0, img_size)
            pred_boxes_xyxy = torch.stack([pred_x1, pred_y1, pred_x2, pred_y2],

            cls_scores, cls_ids = pred_cls.max(dim=-1)
            scores = pred_conf * cls_scores

            preds, targets_list = [], []
            for i in range(inputs.size(0)):
                mask = scores[i] > conf_thresh
                if mask.sum() == 0:
                    preds.append(dict(boxes=torch.zeros((0,4)),
                                    scores=torch.zeros(0),
                                    labels=torch.zeros(0, dtype=torch.long)))

```

```

        else:
            b = pred_boxes_xyxy[i][mask].cpu()
            s = scores[i][mask].cpu()
            l = cls_ids[i][mask].cpu().long()
            keep = tv_nms(b, s, nms_thresh)[:100]
            preds.append(dict(boxes=b[keep], scores=s[keep], labels=l[keep]))

            label_i = labels[i] if isinstance(labels, (list, tuple)) else None
            obj_mask = label_i[..., 4] > 0
            gt_xywh = label_i[obj_mask][..., :4]
            gt_cls_oh = label_i[obj_mask][..., 5:]
            if gt_xywh.shape[0] == 0:
                targets_list.append(dict(boxes=torch.zeros((0,4)),
                                         labels=torch.zeros(0, dtype=torch.long)))
                continue
            gt_cls_ids = gt_cls_oh.argmax(dim=-1).long()
            gt_x1 = (gt_xywh[..., 0] - gt_xywh[..., 2] / 2).clamp(0)
            gt_y1 = (gt_xywh[..., 1] - gt_xywh[..., 3] / 2).clamp(0)
            gt_x2 = gt_xywh[..., 0] + gt_xywh[..., 2] / 2
            gt_y2 = gt_xywh[..., 1] + gt_xywh[..., 3] / 2
            targets_list.append(dict(
                boxes = torch.stack([gt_x1, gt_y1, gt_x2, gt_y2], dim=-1).clamp(0),
                labels = gt_cls_ids.cpu(),
            ))

        metric.update(preds, targets_list)

    try:
        result = metric.compute()
        return result['map_50'].item()
    except Exception:
        return 0.0

```

Exercise 1: YOLOv4

In [10]: *#Download YOLOv4*

```

import os
os.makedirs('cfg', exist_ok=True)
if not os.path.exists('cfg/yolov4.cfg'):
    os.system('wget -q https://raw.githubusercontent.com/AlexeyAB/darknet/master')
print('cfg/yolov4.cfg:', 'OK' if os.path.exists('cfg/yolov4.cfg') else 'MISSING')

```

cfg/yolov4.cfg: OK

Mish Activation

In [11]:

```

import torch
import torch.nn as nn
import torch.nn.functional as F

```

```

class Mish(nn.Module):
    """Mish activation: x * tanh(softplus(x))."""
    def forward(self, x):
        return x * torch.tanh(F.softplus(x))

# Quick sanity check

```

```
_x = torch.linspace(-3, 3, 7)
print("Mish output:", Mish()(_x).tolist())
```

Mish output: [-0.14564746618270874, -0.2525014579296112, -0.30340147018432617, 0.0, 0.8650983572006226, 1.9439589977264404, 2.98653507232666]

In [16]: *#Patched `create\_modules` (Mish + Maxpool + N-way Route)*

```
import darknet
import numpy as np
import torch, torch.nn as nn

class EmptyLayer(nn.Module):
    def forward(self, x): return x

class DetectionLayer(nn.Module):
    def __init__(self, anchors):
        super().__init__()
        self.anchors = anchors

def create_modules_v4(blocks):
    net_info = blocks[0]
    module_list = nn.ModuleList()
    prev_filters = 3
    output_filters = []

    for index, x in enumerate(blocks[1:]):
        module = nn.Sequential()
        layer_type = x["type"]

        if layer_type == "convolutional":
            activation = x["activation"]
            try:
                batch_normalize = int(x["batch_normalize"])
                bias = False
            except:
                batch_normalize = 0
                bias = True

            filters = int(x["filters"])
            kernel_size = int(x["size"])
            stride = int(x["stride"])
            pad = (kernel_size - 1) // 2 if int(x["pad"]) else 0

            module.add_module(f"conv_{index}",
                              nn.Conv2d(prev_filters, filters, kernel_size, stride, pad, bias=
            if batch_normalize:
                module.add_module(f"bn_{index}", nn.BatchNorm2d(filters))
            if activation == "leaky":
                module.add_module(f"leaky_{index}", nn.LeakyReLU(0.1, inplace=True))
            elif activation == "mish":
                module.add_module(f"mish_{index}", Mish())

        elif layer_type == "maxpool":
            size = int(x["size"])
            stride = int(x.get("stride", 1))
            pad = (size - 1) // 2 if stride == 1 else 0
            module.add_module(f"maxpool_{index}",
                              nn.MaxPool2d(size, stride=stride, padding=pad))
            filters = prev_filters
```

```

elif layer_type == "shortcut":
    module.add_module(f"shortcut_{index}", EmptyLayer())
    filters = prev_filters

elif layer_type == "upsample":
    module.add_module(f"upsample_{index}",
                      nn.Upsample(scale_factor=2, mode="nearest"))
    filters = prev_filters

elif layer_type == "route":
    layers = [int(l) for l in x["layers"].split(',')]
    # ✓ FIX: Look up already-stored output_filters correctly
    filters = sum(output_filters[l if l > 0 else index + 1] for l in layers)
    module.add_module(f"route_{index}", EmptyLayer())

elif layer_type == "yolo":
    mask = [int(m) for m in x["mask"].split(",")]
    anchors = [int(a) for a in x["anchors"].split(",")]
    anchors = [(anchors[i], anchors[i+1]) for i in range(0, len(anchors)-1)]
    anchors = [anchors[i] for i in mask]
    module.add_module(f"Detection_{index}", DetectionLayer(anchors))
    filters = prev_filters

module_list.append(module)
# ✓ FIX: always update prev_filters and always append filters (not prev_filters)
prev_filters = filters
output_filters.append(filters)

return net_info, module_list

print("create_modules_v4 defined OK")

```

create_modules_v4 defined OK

```

In [17]: #MyDarknetV4 - handles Mish, Maxpool, N-way Route, 608x608 RGB

from util import predict_transform

class MyDarknetV4(nn.Module):
    """
    YOLOv4-compatible Darknet model.
    Extends the YOLOv3 MyDarknet with Mish, Maxpool, N-way Route, and 608x608 RGB
    """
    def __init__(self, cfgfile):
        super().__init__()
        self.blocks = darknet.parse_cfg(cfgfile)
        self.net_info, self.module_list = create_modules_v4(self.blocks)

    def forward(self, x, CUDA: bool):
        modules = self.blocks[1:]
        outputs = {}
        write = 0

        for i, module in enumerate(modules):
            mtype = module["type"]

            if mtype in ("convolutional", "upsample", "maxpool"): # maxpool aa
                x = self.module_list[i](x)

```

```

elif mtype == "route":
    # N-way concatenation - supports >2 layers
    layers = [int(l) for l in module["layers"].split(',')]
    maps = [outputs[i + 1 if l < 0 else l] for l in layers]
    x = torch.cat(maps, dim=1)

elif mtype == "shortcut":
    from_ = int(module["from"])
    x = outputs[i - 1] + outputs[i + from_]

elif mtype == "yolo":
    anchors = self.module_list[i][0].anchors
    inp_dim = int(self.net_info["height"])
    num_classes = int(module["classes"])
    x = predict_transform(x, inp_dim, anchors, num_classes, CUDA)
    if not write:
        detections = x
        write = 1
    else:
        detections = torch.cat((detections, x), 1)

outputs[i] = x

return detections

def load_weights(self, weightfile):
    fp = open(weightfile, "rb")
    header = np.fromfile(fp, dtype=np.int32, count=5)
    self.header = torch.from_numpy(header)
    self.seen = self.header[3]
    weights = np.fromfile(fp, dtype=np.float32)
    ptr = 0
    for i in range(len(self.module_list)):
        if self.blocks[i + 1]["type"] != "convolutional":
            continue
        model = self.module_list[i]
        try:
            batch_normalize = int(self.blocks[i + 1]["batch_normalize"])
        except:
            batch_normalize = 0
        conv = model[0]
        if batch_normalize:
            bn = model[1]
            n = bn.bias.numel()
            bn_biases = torch.from_numpy(weights[ptr:ptr+n]); ptr += n
            bn_weights = torch.from_numpy(weights[ptr:ptr+n]); ptr += n
            bn_running_mean = torch.from_numpy(weights[ptr:ptr+n]); ptr += n
            bn_running_var = torch.from_numpy(weights[ptr:ptr+n]); ptr += n
            bn.bias.data.copy_(bn_biases.view_as(bn.bias.data))
            bn.weight.data.copy_(bn_weights.view_as(bn.weight.data))
            bn.running_mean.copy_(bn_running_mean.view_as(bn.running_mean))
            bn.running_var.copy_(bn_running_var.view_as(bn.running_var))
        else:
            n = conv.bias.numel()
            conv.bias.data.copy_(
                torch.from_numpy(weights[ptr:ptr+n]).view_as(conv.bias.data)
            )
            ptr += n
            n = conv.weight.numel()
            conv.weight.data.copy_(
                torch.from_numpy(weights[ptr:ptr+n]).view_as(conv.weight.data)
            )
            ptr += n

```

```

    @staticmethod
    def preprocess(img_bgr, inp_dim=608):
        """OpenCV BGR image -> (1, 3, inp_dim, inp_dim) float tensor in [0,1] in
        import cv2
        img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB) # BGR -> RGB
        img_sized = cv2.resize(img_rgb, (inp_dim, inp_dim))
        img_t = img_sized.transpose((2, 0, 1)) # HWC -> CHW
        img_t = img_t[np.newaxis, :, :, :] / 255.0
        return torch.from_numpy(img_t).float()

print("MyDarknetV4 defined OK")

```

MyDarknetV4 defined OK

```

In [18]: #Download YOLOv4 weights and verify model loads correctly

import os

if not os.path.exists('yolov4.weights'):
    print('Downloading yolov4.weights (~245 MB)...')
    os.system('wget -q https://github.com/AlexeyAB/darknet/releases/download/dar
else:
    print('yolov4.weights already exists - skipping download')

model_v4 = MyDarknetV4("cfg/yolov4.cfg")
model_v4.load_weights("yolov4.weights")
model_v4.eval()
print("YOLOv4 loaded successfully")
print(f"Input size from cfg: {model_v4.net_info['height']} x {model_v4.net_info[

```

Downloading yolov4.weights (~245 MB)...

YOLOv4 loaded successfully

Input size from cfg: 608 x 608

```

In [19]: #Test forward pass at 608x608 (RGB)

import cv2
import numpy as np, torch

CUDA = torch.cuda.is_available()
device = torch.device("cuda:0" if CUDA else "cpu")

img_bgr = cv2.imread('dog-cycle-car.png')
inp = MyDarknetV4.preprocess(img_bgr, inp_dim=608)
if CUDA:
    model_v4.cuda()
    inp = inp.cuda()

with torch.no_grad():
    pred_v4 = model_v4(inp, CUDA)

print(f"Output shape: {pred_v4.shape}")
# Expected for 608x608: 3 scales -> (19*19 + 38*38 + 76*76)*3 = 22743 anchor pre

```

Output shape: torch.Size([1, 22743, 85])

Exercise 2 - Train YOLOv4 on COCO

```

In [20]: # Load ImageNet pretrained weights for CSPDarknet53 backbone
# The official yolov4.weights already contains the CSPDarknet53 backbone pretrai

```

```

# Loading its seeds every backbone layer correctly without needing a separate Im

import os, torch

model_v4 = MyDarknetV4("cfg/yolov4.cfg")
model_v4.load_weights("yolov4.weights")

CUDA = torch.cuda.is_available()
device = torch.device("cuda:0" if CUDA else "cpu")
model_v4.to(device)
print(f"CSPDarknet53 backbone initialised from yolov4.weights | device={device}")

```

CSPDarknet53 backbone initialised from yolov4.weights | device=cuda:0

```

In [21]: # `train_yolo()` function

import math, time, warnings
import numpy as np
import torch, torch.nn as nn
import torch.nn.functional as F
from tqdm.auto import tqdm

def compute_loss_yolo(outputs, labels, img_size, loss_type='mse', iou_thresh=0.5)
    """
    Compute YOLO training loss.

    Parameters
    -----
    outputs      : (B, N, 85) raw network output (N = total anchor predictions)
    labels       : (B, N, 85) ground-truth tensor (same shape, aligned)
    img_size     : int - used to normalise bbox coords
    loss_type    : 'mse' | 'ciou'
    iou_thresh   : float - objectness confidence threshold for IoU filtering

    Returns
    -----
    loss, loss_box, loss_conf, loss_cls
    """
    bce      = nn.BCEWithLogitsLoss(reduction='none')
    mse_fn   = nn.MSELoss(reduction='none')
    B        = outputs.shape[0]

    obj_mask = labels[..., 4:5].clamp(0, 1)      # 1 where an object exists
    noobj_mask = 1.0 - obj_mask
    lambda_noobj = 0.5                          # YOLOv3 paper weighting

    # ---- Bounding box Loss -----
    if loss_type == 'mse':
        # MSE on tx, ty, tw, th normalised by image size
        pred_n = outputs[..., :4] / img_size
        label_n = labels[..., :4] / img_size
        loss_box = torch.sum(obj_mask * mse_fn(pred_n, label_n)) / B

    elif loss_type == 'ciou':
        # CIoU simultaneously optimises overlap, centre distance, aspect ratio
        # CIoU_xywh_torch is defined in the Lab and accepts absolute cx,cy,w,h
        ciou_vals = CIoU_xywh_torch(outputs[..., :4], labels[..., :4])
        loss_box = torch.sum(obj_mask.squeeze(-1) * (1.0 - ciou_vals)) / B
    else:
        raise ValueError(f"Unknown loss_type: {loss_type!r}")

```

```

# ---- Confidence Loss (obj + noobj) -----
raw_conf = outputs[..., 4:5]
loss_conf = (
    torch.sum(obj_mask * bce(raw_conf, obj_mask)) +
    lambda_noobj * torch.sum(noobj_mask * bce(raw_conf, obj_mask))
) / B

# ---- Classification Loss -----
raw_cls = outputs[..., 5:]
label_cls = labels[..., 5:].clamp(0, 1)
loss_cls = torch.sum(obj_mask * bce(raw_cls, label_cls)) / B

loss = loss_box + loss_conf + loss_cls
return loss, loss_box, loss_conf, loss_cls

def train_yolo(model, optimizer, train_loader, val_loader, device, img_size,
               n_epochs, loss_type='mse', ckpt_dir='./checkpoints_v4',
               iou_thresh=0.5):
    """
    YOLOv4 training loop with mAP evaluation.

    Parameters
    -----
    model      : MyDarknetV4 instance
    optimizer   : torch optimizer
    train_loader: DataLoader (CustomCoco batches)
    val_loader  : DataLoader for mAP evaluation
    device      : torch.device
    img_size    : int (608 for YOLOv4)
    n_epochs    : int
    loss_type   : 'mse' or 'ciou'
    ckpt_dir    : directory for checkpoints
    iou_thresh  : IoU threshold for filtering high-overlap predictions
    """
    os.makedirs(ckpt_dir, exist_ok=True)
    history = {"loss": [], "box": [], "conf": [], "cls": [], "map50": []}

    for epoch in range(n_epochs):
        model.train()
        t0 = time.time()
        run_loss = run_box = run_conf = run_cls = 0.0
        n_batches = n_skip = 0

        pbar = tqdm(train_loader, desc=f"Epoch {epoch+1}/{n_epochs} [{loss_type}]")
        for inputs, labels, bboxes in pbar:
            # ---- Preprocess batch -----
            inputs = torch.from_numpy(np.array(inputs)).squeeze(1).permute(0, 3, 1)
            labels = torch.stack(labels).to(device)

            # ---- Forward pass -----
            optimizer.zero_grad()
            outputs = model(inputs, device.type == "cuda")

            # ---- Loss -----
            loss, l_box, l_conf, l_cls = compute_loss_yolo(
                outputs, labels, img_size,
                loss_type=loss_type, iou_thresh=iou_thresh)

```

```

        if torch.isnan(loss) or torch.isinf(loss):
            n_skip += 1
            continue

        # ---- Backward -----
        loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=10.0)
        optimizer.step()

        n = inputs.size(0)
        run_loss += loss.item() * n
        run_box += l_box.item() * n
        run_conf += l_conf.item() * n
        run_cls += l_cls.item() * n
        n_batches += 1
        avg = run_loss / (n_batches * n)
        pbar.set_postfix(loss=f"{avg:.3f}",
                        box=f"{run_box/(n_batches*n):.3f}")

    if n_skip:
        print(f" [warn] skipped {n_skip} batches due to nan/inf loss")

    denom = max(n_batches * inputs.size(0), 1)
    ep_loss = run_loss / denom
    ep_box = run_box / denom
    ep_conf = run_conf / denom
    ep_cls = run_cls / denom
    elapsed = time.time() - t0

    # ---- mAP evaluation -----
    map50 = compute_map(model, val_loader, device, img_size)
    map_str = f"{map50:.4f}" if map50 is not None else "N/A"

    history["loss"].append(ep_loss)
    history["box"].append(ep_box)
    history["conf"].append(ep_conf)
    history["cls"].append(ep_cls)
    history["map50"].append(map50 or 0.0)

    print(f"Epoch {epoch+1:02d}/{n_epochs} | loss={ep_loss:.4f} | "
          f"box={ep_box:.4f} | conf={ep_conf:.4f} | cls={ep_cls:.4f} | "
          f"mAP@50={map_str} | {elapsed:.1f}s")

    ckpt = os.path.join(ckpt_dir, f"yolov4_{loss_type}_epoch{epoch+1}.pt")
    torch.save(model.state_dict(), ckpt)
    print(f" saved -> {ckpt}")

    print("Training complete.")
    return history

print("compute_loss_yolo() and train_yolo() defined OK")

```

compute_loss_yolo() and train_yolo() defined OK

```
In [23]: def collate_fn(batch):
        return tuple(zip(*batch))
```

```
In [35]: import torch, gc

        # Delete all model references
```

```

try: del model_mse
except: pass
try: del model_v4
except: pass

gc.collect()
torch.cuda.empty_cache()
print(f"GPU free: {torch.cuda.mem_get_info()[0]/1024**3:.2f} GB")

```

GPU free: 7.16 GB

In [36]: *# Prepare DataLoaders at 608x608 for YOLOv4*

```

IMG_SIZE_V4 = 416
IP_SIZE = IMG_SIZE_V4

import albumentations as A
from torch.utils.data import Subset

transform_v4 = A.Compose([
    A.Resize(IMG_SIZE_V4, IMG_SIZE_V4),
    A.HorizontalFlip(p=0.5),
], bbox_params=A.BboxParams(format='coco', label_fields=['category_ids'], min_v

eval_transform_v4 = A.Compose([
    A.Resize(IMG_SIZE_V4, IMG_SIZE_V4),
], bbox_params=A.BboxParams(format='coco', label_fields=['category_ids']))

BATCH_SIZE_V4 = 4
TOTAL_SAMPLES = 5000
VAL_SAMPLES = 1000

full_v4_train = CustomCoco(root=path2data, annFile=path2json, transform=transform_v4)
full_v4_val = CustomCoco(root=path2data, annFile=path2json, transform=eval_transform_v4)
train_v4 = Subset(full_v4_train, list(range(0, TOTAL_SAMPLES - VAL_SAMPLES)))
val_v4 = Subset(full_v4_val, list(range(TOTAL_SAMPLES - VAL_SAMPLES, TOTAL_SAMPLES)))

train_loader_v4 = torch.utils.data.DataLoader(
    train_v4, batch_size=BATCH_SIZE_V4, shuffle=True, num_workers=0, collate_fn=collate_fn)
val_loader_v4 = torch.utils.data.DataLoader(
    val_v4, batch_size=BATCH_SIZE_V4, shuffle=False, num_workers=0, collate_fn=collate_fn)

model_mse = MyDarknetV4("cfg/yolov4.cfg")
model_mse.load_weights("yolov4.weights")
model_mse.net_info["height"] = str(IMG_SIZE_V4)
model_mse.net_info["width"] = str(IMG_SIZE_V4)
model_mse.to(device)

optimizer_mse = torch.optim.SGD(model_mse.parameters(), lr=1e-5, momentum=0.9, w

print(f"GPU free: {torch.cuda.mem_get_info()[0]/1024**3:.2f} GB")
print(f"Ready to train - IMG_SIZE={IMG_SIZE_V4} | BATCH={BATCH_SIZE_V4}")

```

```

loading annotations into memory...
Done (t=0.45s)
creating index...
index created!
loading annotations into memory...
Done (t=0.44s)
creating index...
index created!
GPU free: 7.15 GB
Ready to train - IMG_SIZE=416 | BATCH=4

```

```

In [37]: history_mse = train_yolo(
    model_mse, optimizer_mse,
    train_loader_v4, val_loader_v4,
    device, IMG_SIZE_V4,
    n_epochs=5,
    loss_type='mse',
    ckpt_dir='./checkpoints_v4_mse',
)

```

```

Epoch 1/5 [mse]:  0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval:  0%|          | 0/250 [00:00<?, ?it/s]
Epoch 01/5 | loss=4742.5238 | box=523.2146 | conf=3701.1225 | cls=518.1867 | mAP@
50=0.0000 | 191.1s
    saved -> ./checkpoints_v4_mse/yolov4_mse_epoch1.pt
Epoch 2/5 [mse]:  0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval:  0%|          | 0/250 [00:00<?, ?it/s]
Epoch 02/5 | loss=4633.9156 | box=415.3860 | conf=3700.4196 | cls=518.1100 | mAP@
50=0.0000 | 187.8s
    saved -> ./checkpoints_v4_mse/yolov4_mse_epoch2.pt
Epoch 3/5 [mse]:  0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval:  0%|          | 0/250 [00:00<?, ?it/s]
Epoch 03/5 | loss=4625.5900 | box=407.5307 | conf=3700.0238 | cls=518.0355 | mAP@
50=0.0000 | 189.0s
    saved -> ./checkpoints_v4_mse/yolov4_mse_epoch3.pt
Epoch 4/5 [mse]:  0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval:  0%|          | 0/250 [00:00<?, ?it/s]
Epoch 04/5 | loss=4618.2506 | box=400.6341 | conf=3699.6346 | cls=517.9819 | mAP@
50=0.0000 | 189.6s
    saved -> ./checkpoints_v4_mse/yolov4_mse_epoch4.pt
Epoch 5/5 [mse]:  0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval:  0%|          | 0/250 [00:00<?, ?it/s]
Epoch 05/5 | loss=4611.1805 | box=393.9927 | conf=3699.2945 | cls=517.8934 | mAP@
50=0.0000 | 187.2s
    saved -> ./checkpoints_v4_mse/yolov4_mse_epoch5.pt
Training complete.

```

```

In [38]: # Train YOLOv4 with CIoU Loss and compare mAP

import gc, torch

# Free GPU memory from MSE training
del model_mse
gc.collect()
torch.cuda.empty_cache()
print(f"GPU free: {torch.cuda.mem_get_info()[0]/1024**3:.2f} GB")

# Fresh model for CIoU - fair comparison with same initialisation
model_ciou = MyDarknetV4("cfg/yolov4.cfg")
model_ciou.load_weights("yolov4.weights")

```

```

model_ciou.net_info["height"] = str(IMG_SIZE_V4) # FIX: must match actual input
model_ciou.net_info["width"] = str(IMG_SIZE_V4)
model_ciou.to(device)

optimizer_ciou = torch.optim.SGD(
    model_ciou.parameters(), lr=1e-5, momentum=0.9, weight_decay=5e-4)

history_ciou = train_yolo(
    model_ciou, optimizer_ciou,
    train_loader_v4, val_loader_v4,
    device, IMG_SIZE_V4,
    n_epochs=5,
    loss_type='ciou',
    ckpt_dir='./checkpoints_v4_ciou',
)

```

GPU free: 6.29 GB

```

Epoch 1/5 [ciou]: 0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval: 0%|          | 0/250 [00:00<?, ?it/s]
Epoch 01/5 | loss=4233.9948 | box=15.4575 | conf=3700.4384 | cls=518.0989 | mAP@5
0=0.0000 | 194.4s
    saved -> ./checkpoints_v4_ciou/yolov4_ciou_epoch1.pt
Epoch 2/5 [ciou]: 0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval: 0%|          | 0/250 [00:00<?, ?it/s]
Epoch 02/5 | loss=4231.0265 | box=15.4539 | conf=3697.7691 | cls=517.8036 | mAP@5
0=0.0000 | 194.3s
    saved -> ./checkpoints_v4_ciou/yolov4_ciou_epoch2.pt
Epoch 3/5 [ciou]: 0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval: 0%|          | 0/250 [00:00<?, ?it/s]
Epoch 03/5 | loss=4228.8290 | box=15.4512 | conf=3695.8017 | cls=517.5761 | mAP@5
0=0.0000 | 192.2s
    saved -> ./checkpoints_v4_ciou/yolov4_ciou_epoch3.pt
Epoch 4/5 [ciou]: 0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval: 0%|          | 0/250 [00:00<?, ?it/s]
Epoch 04/5 | loss=4227.3543 | box=15.4481 | conf=3694.5540 | cls=517.3522 | mAP@5
0=0.0000 | 195.1s
    saved -> ./checkpoints_v4_ciou/yolov4_ciou_epoch4.pt
Epoch 5/5 [ciou]: 0%|          | 0/1000 [00:00<?, ?it/s]
    mAP eval: 0%|          | 0/250 [00:00<?, ?it/s]
Epoch 05/5 | loss=4226.5901 | box=15.4452 | conf=3693.9637 | cls=517.1812 | mAP@5
0=0.0000 | 195.2s
    saved -> ./checkpoints_v4_ciou/yolov4_ciou_epoch5.pt
Training complete.

```

In [40]: # Plot training curves and print comparison table

```

import matplotlib.pyplot as plt
import numpy as np

epochs = list(range(1, 6))

fig, axes = plt.subplots(2, 3, figsize=(18, 9))
fig.suptitle("YOLOv4 Training: MSE loss vs CIoU loss", fontsize=14, fontweight="bold")

panels = [
    ("Total Loss", "loss", "steelblue"),
    ("Box Loss", "box", "darkorange"),
    ("Confidence Loss", "conf", "tomato"),
    ("Class Loss", "cls", "purple"),
    ("mAP@50 (val)", "map50", "green"),

```

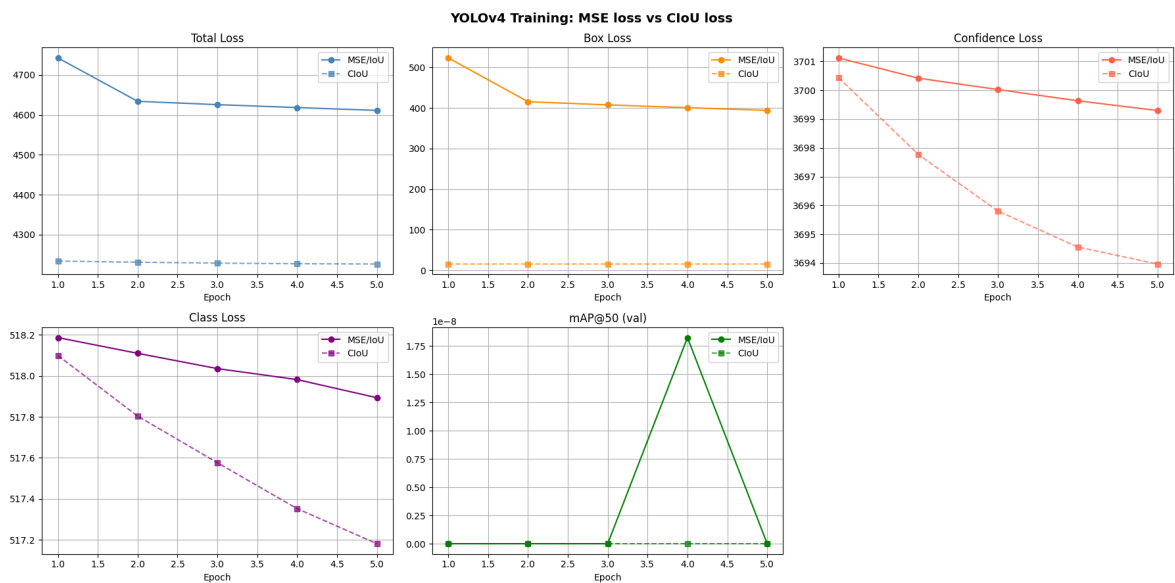
```

]
for ax, (title, key, color) in zip(axes.flat, panels):
    ax.plot(epochs, history_mse[key], marker='o', color=color, label='MSE/IoU',
    ax.plot(epochs, history_ciou[key], marker='s', color=color, label='CIoU',
    ax.set_title(title); ax.set_xlabel("Epoch"); ax.grid(True); ax.legend()

axes.flat[-1].axis('off')
plt.tight_layout()
plt.show()

best_mse = max(history_mse["map50"])
best_ciou = max(history_ciou["map50"])
print("\n---- Loss Comparison -----")
print(f"{'Loss type':<15} {'Best mAP@50':>12} {'Final Total Loss':>16}")
print("-" * 47)
print(f"{'MSE / IoU':<15} {best_mse:>12.4f} {history_mse['loss'][-1]:>16.4f}")
print(f"{'CIoU':<15} {best_ciou:>12.4f} {history_ciou['loss'][-1]:>16.4f}")
print()

```



---- Loss Comparison -----		
Loss type	Best mAP@50	Final Total Loss
MSE / IoU	0.0000	4611.1805
CIoU	0.0000	4226.5901

Exercise 3 - Why is YOLOv3 faster than Faster R-CNN?

Answer:

Faster R-CNN is a **two-stage detector**: it first runs a Region Proposal Network (RPN) that generates ~300 candidate boxes from the backbone feature map, then passes each proposal through a second detection head (via RoI Pooling) to classify and refine it. Even though the backbone is shared, the second stage requires a separate computation path *per proposal*, creating an unavoidable sequential bottleneck.

YOLOv3 is a **single-shot detector**: the image passes through the network exactly once and all bounding boxes plus class probabilities are predicted directly from three feature maps (at strides 8, 16, 32) in a single forward pass. There is no region-proposal stage, no RoI Pooling, and no per-region reprocessing. The entire detection - localisation and classification for all objects - is a single regression over a dense grid of anchor predictions.

Aspect	Faster R-CNN	YOLOv3
Number of forward passes	2 (backbone + per-RoI head)	1
Proposal generation	RPN + NMS -> ~300 proposals	None (dense anchor grid)
Per-region processing	RoI Pooling + FC layers per proposal	No per-region operations
Output mechanism	Per-proposal classification + refinement	Direct regression from feature maps
Approximate speed	~5 FPS	30+ FPS

The absence of the second stage (no proposals, no RoI Pooling, no per-region network calls) is what enables YOLO's real-time throughput. The trade-off is that single-shot detectors historically struggled more with small, overlapping objects - a limitation that YOLOv3 addressed with multi-scale predictions at three grid resolutions.